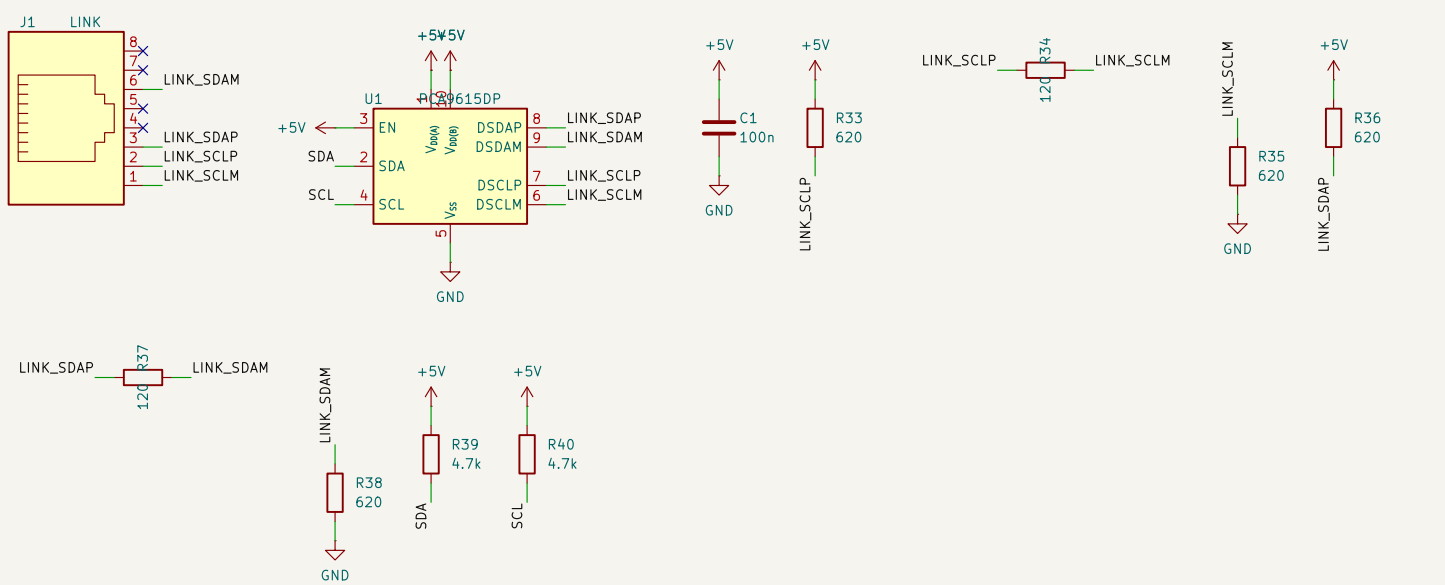
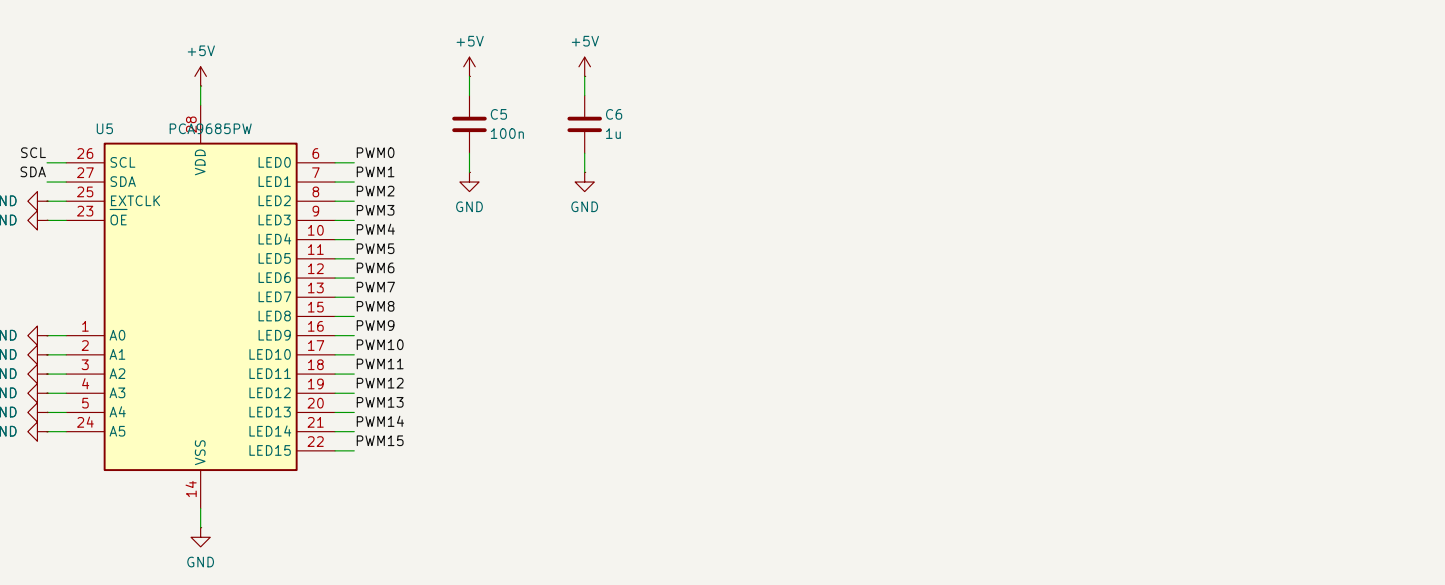


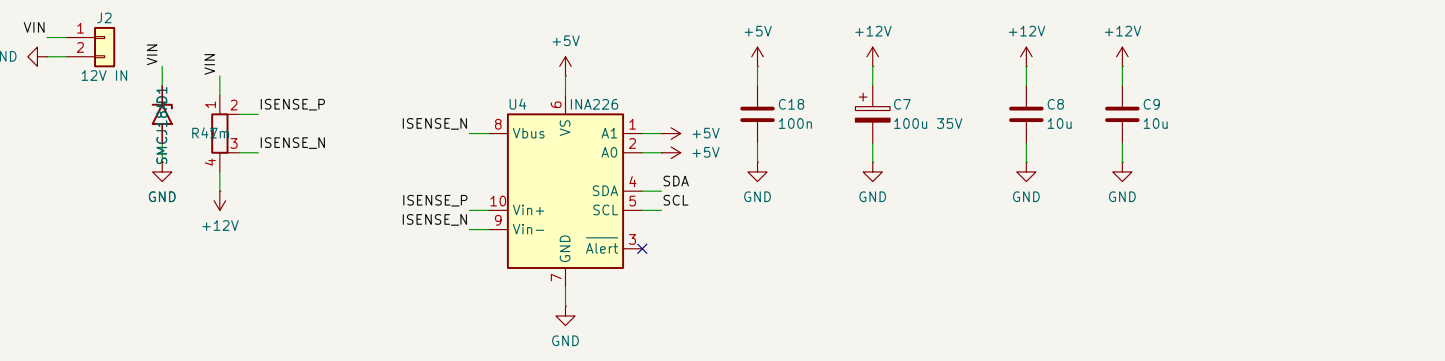
LINK: RJ45 to the sensor board's J6, differential I2C (PCA9615)



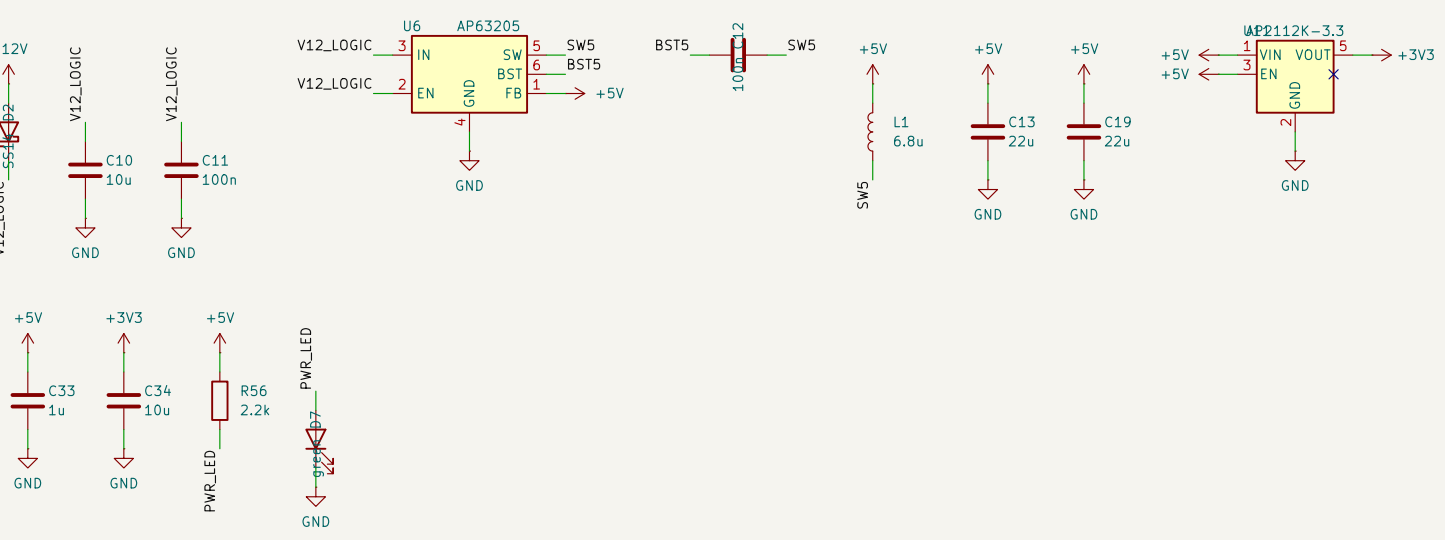
PWM: PCA9685 at 0x40, 16 channels



12 V IN (20 A) and its current monitor: INA226 at 0x45



LOGIC SUPPLIES: 5 V buck, 3.3 V for the RP2040

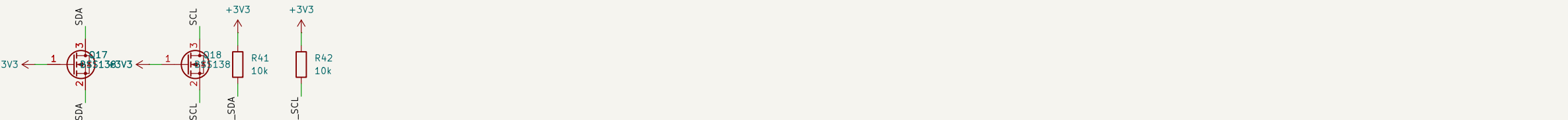
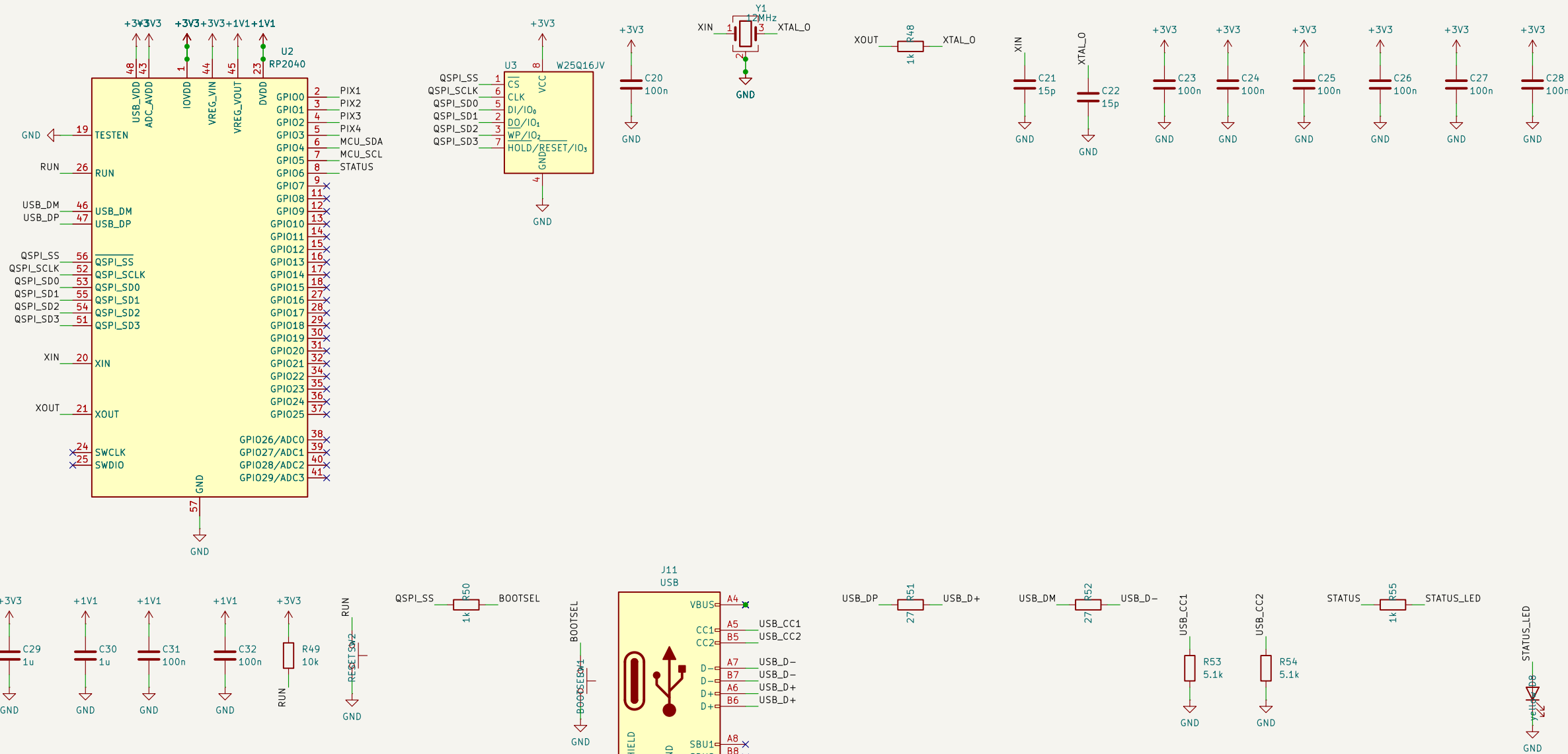


The link: any straight-through Cat5e/Cat6 patch cable to the sensor board's J6 -- NOT Ethernet: never plug either end into a switch or a PoE injector. Pairs 1/2 (SCL) and 3/6 (SDA) carry the I2C bus as differential I2C. 4/5 and 7/8 are reserved. Each pair is terminated at both ends. 620/120/620 ohm. No ground in the cable: the strips' current returns on the 12 V input's own ground wire, never through the link or the Pi.
The bus: PCA9685 0x40, INA226 0x45, RP2040 0x30. Keep it at 100 kHz.

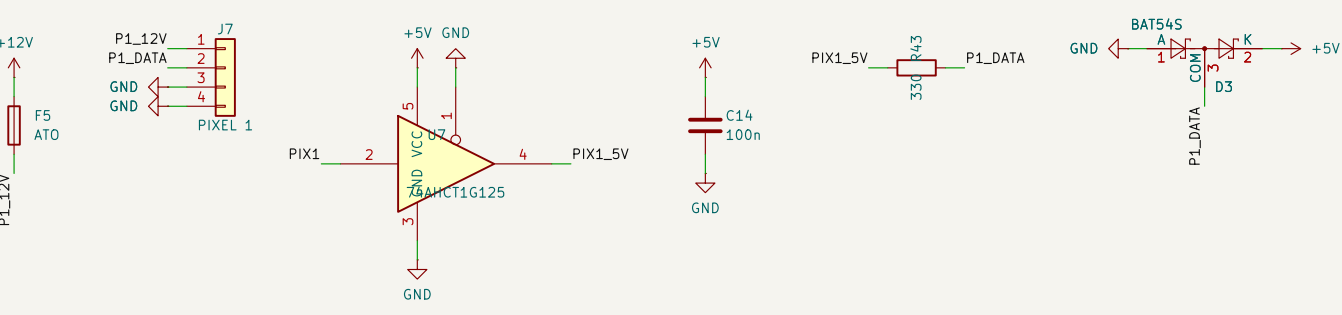
Fuses: F1-F8 are standard blade (ATO/ATC) fuses, one per output, sized for that output's wire -- 10 A for 5 m of 12 V strip on 16 AWG. The whole board is 20 A. J2 takes 10 AWG from a 25 A fuse or breaker at the supply, and the software holds the total under 20 A with the INA226's reading (everything at full white would draw about 60 A).

Zones: common-anode 12 V RGBW strips; zone 2 colour c is PCA9685 output 4 (c=1) + (R 0, G 1, B 2, W 3). Pixels: 12 V strips (WS2815...); B1 is a WS2815's backup data input; grounded here for the strip's first pixel. Firmware: hold BOOTSEL, press RESET, and the RP2040 appears on a computer (USB-C, J11) as a drive: copy the firmware's .uf2 onto it. The board must have its 12 V while it's plugged in; J11's VBUS is unused.

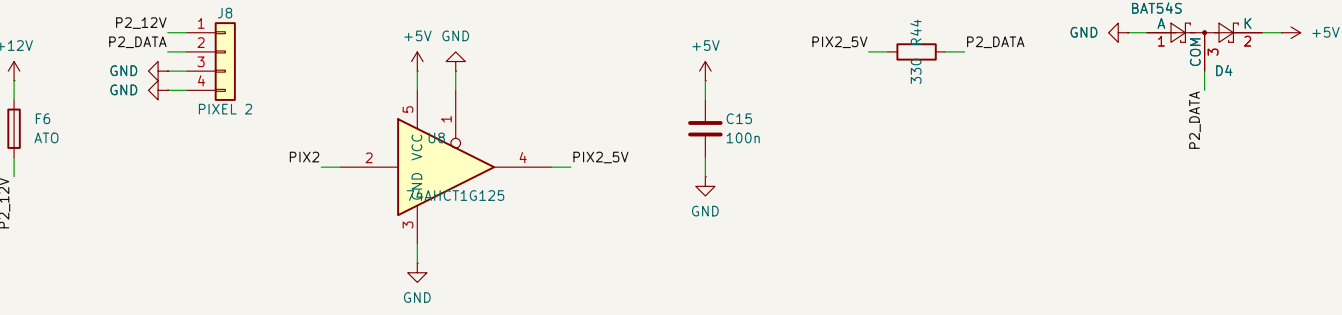
RP2040: the addressable outputs, I2C target at 0x30



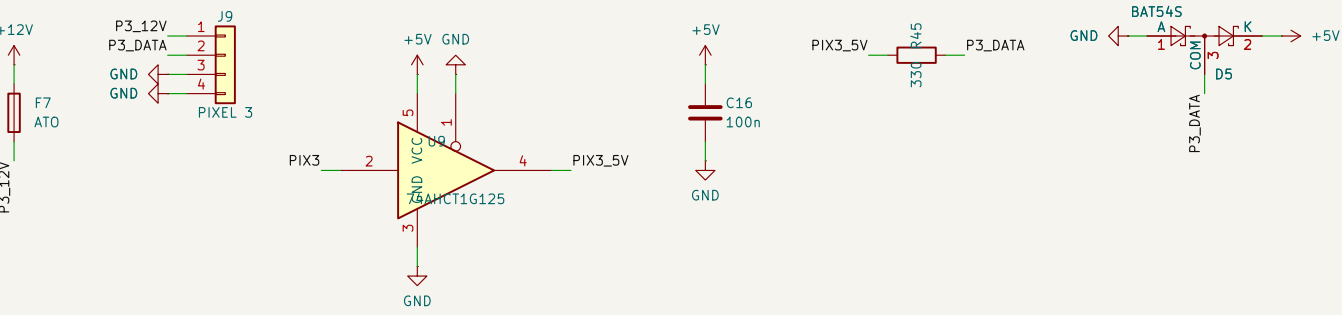
PIXEL 1: 12 V addressable strip, RP2040 GPIO0



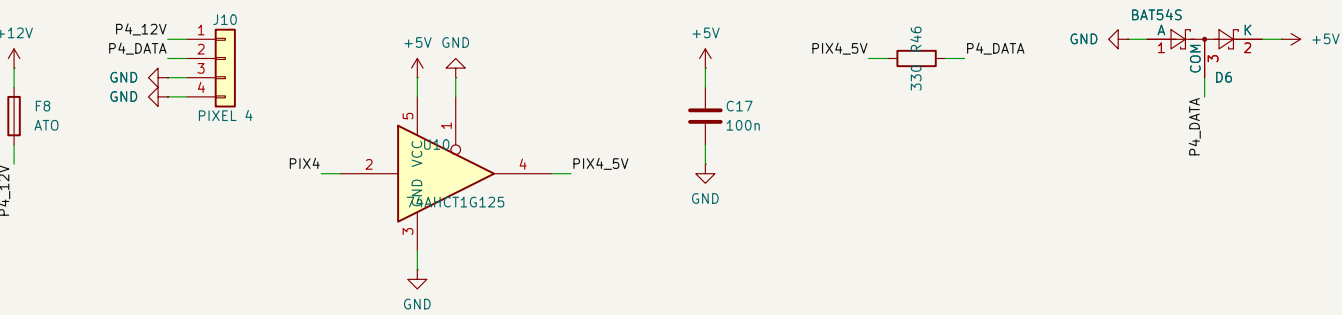
PIXEL 2: 12 V addressable strip, RP2040 GPIO1



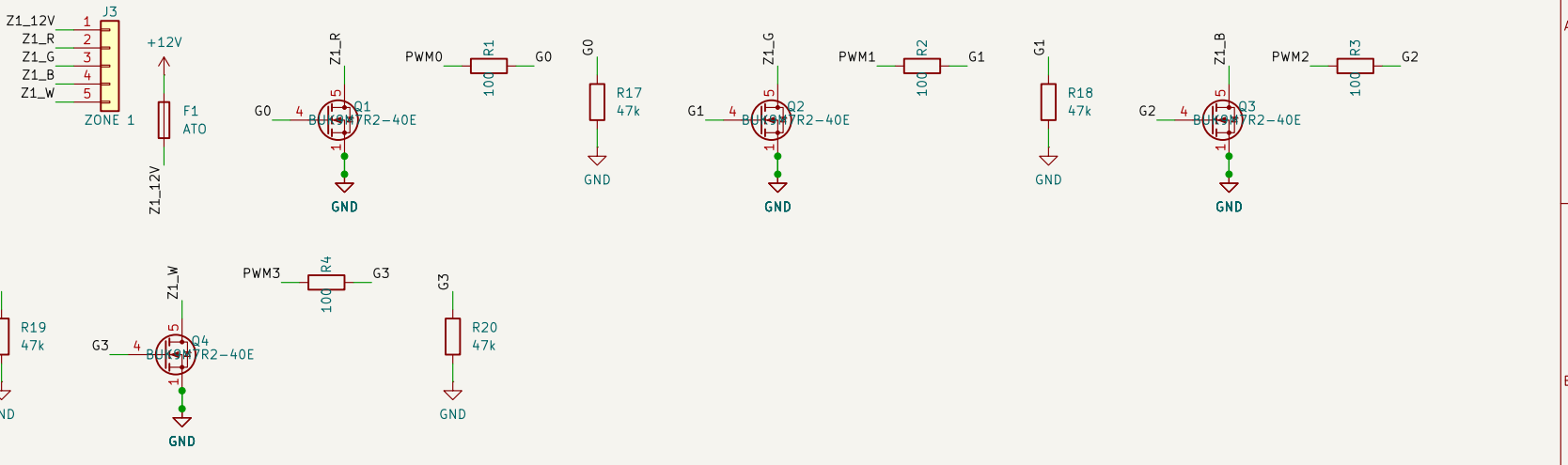
PIXEL 3: 12 V addressable strip, RP2040 GPIO2



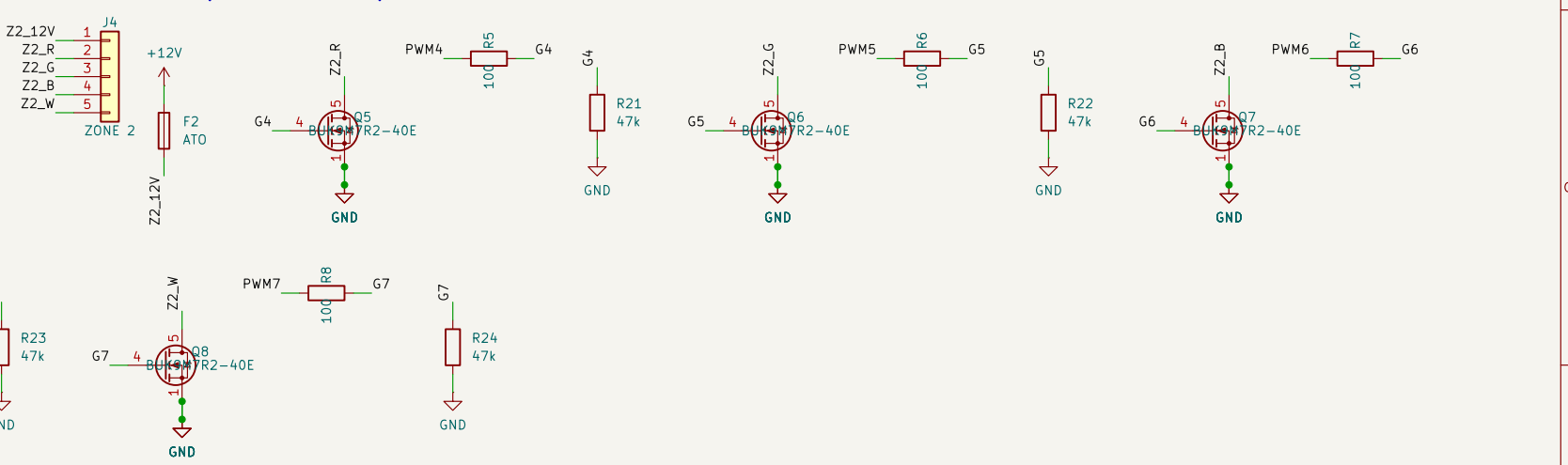
PIXEL 4: 12 V addressable strip, RP2040 GPIO3



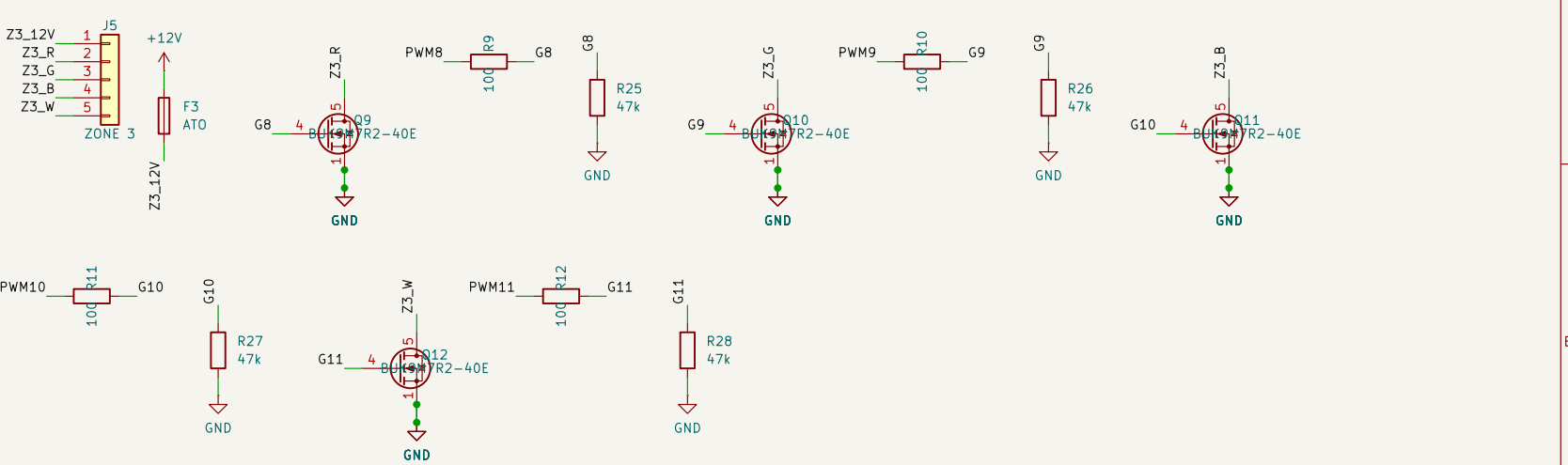
ZONE 1: RGBW strip, PCA9685 outputs 0-3



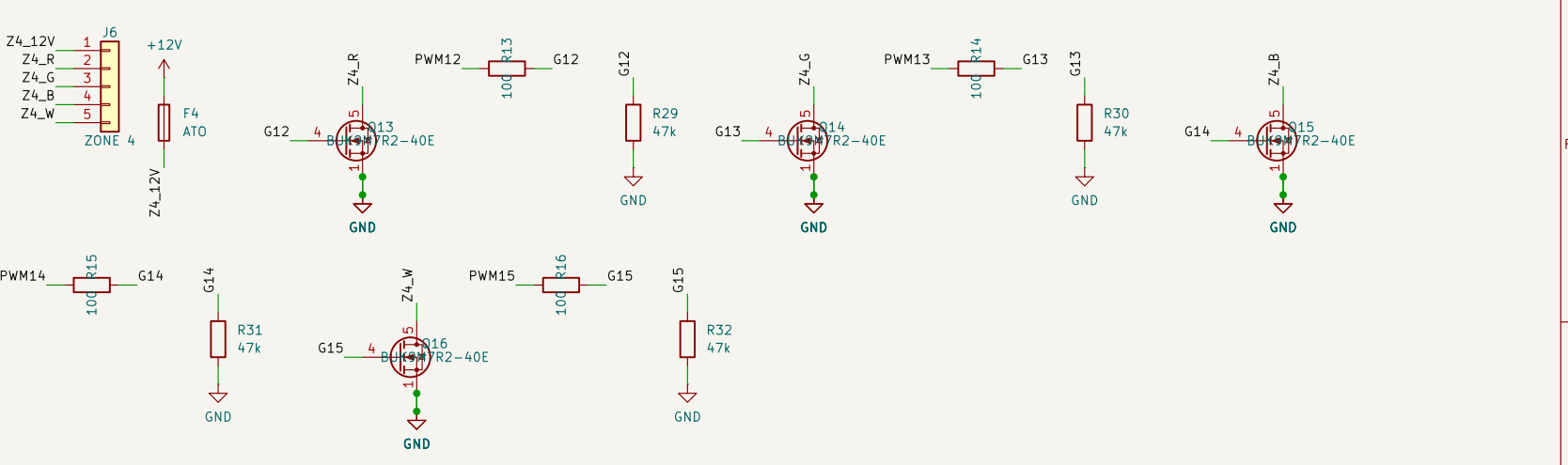
ZONE 2: RGBW strip, PCA9685 outputs 4-7



ZONE 3: RGBW strip, PCA9685 outputs 8-11



ZONE 4: RGBW strip, PCA9685 outputs 12-15



MOUNTING HOLES, POWER FLAGS

